北京大學

第三讲
统计推断方法

机器学习概论

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2025年9月21日星期日

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统计推断方法

- 之前，概率分布的参数是已知的（如高斯分布的均值和方差）。现在讨论在参数未知情况下，如何学习或估计概率参数
- 给定分布函数$p(\boldsymbol{x};\boldsymbol{\theta})$，其中$\boldsymbol{\theta}$是未知参数，以及观测到的数据集$\mathcal{D}$，学习参数$\boldsymbol{\theta}$的过程可视为从数据集$\mathcal{D}$来估计$\boldsymbol{\theta}$的过程。这一过程称为模型拟合 (model fitting)。这是以概率为模型框架的机器学习体系的中心思想
- 对参数$\boldsymbol{\theta}$的估计可分为两大类方法
  - 视$\boldsymbol{\theta}$为未知变量，学习的目标是求解$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, p(\mathcal{D};\boldsymbol{\theta})$
    - ✓频率统计方法 (Frequentist statistics)。典型的方法：最大似然估计
  - 视$\boldsymbol{\theta}$为未知随机变量，学习的目标是求解后验分布$p(\boldsymbol{\theta}|\mathcal{D})$
    - ✓贝叶斯统计方法 (Bayesian statistics)

最大似然估计

- Maximum Likelihood Estimation (MLE)
- 最常用的参数估计方法，形式如下

似然函数

$$\boldsymbol{\theta}_{MLE} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, \boxed{p(\mathcal{D};\boldsymbol{\theta})}$$

  – 其中，$\mathcal{D}$中的数据假定为独立地采样自相同的分布。即$\mathcal{D}$中的数据是独立同分布的 (independent and identically distributed)

- 数据集$\mathcal{D}$有两种形式
  – $\mathcal{D} = \{\boldsymbol{x}_n\}_{n=1}^{N}$： 无标签数据集
    ✓无监督学习
  – $\mathcal{D} = \{(\boldsymbol{x}_n, y_n)\}_{n=1}^{N}$： 有标签数据集，其中$\boldsymbol{x}_n$是输入，$y_n$是对应输出（标签）
    ✓监督学习

最大似然估计

- $\mathcal{D}=\{\boldsymbol{x}_n\}_{n=1}^N$，似然函数

$$p(\mathcal{D};\boldsymbol{\theta})=\prod_{n=1}^{N}p(\boldsymbol{x}_n;\boldsymbol{\theta}) \qquad \boldsymbol{\theta}_{MLE}=\operatorname*{argmax}_{\boldsymbol{\theta}} p(\mathcal{D};\boldsymbol{\theta})$$

- 对数似然 (log likelihood)

$$LL(\boldsymbol{\theta})\triangleq\log p(\mathcal{D};\boldsymbol{\theta})=\sum_{n=1}^{N}\log p(\boldsymbol{x}_n;\boldsymbol{\theta}) \qquad \boldsymbol{\theta}_{MLE}=\operatorname*{argmax}_{\boldsymbol{\theta}} LL(\boldsymbol{\theta})$$

- 负对数似然 (negative log likelihood)
  - 优化问题通常统一转化为最小化问题

$$NLL(\boldsymbol{\theta})\triangleq-\log p(\mathcal{D};\boldsymbol{\theta})=-\sum_{n=1}^{N}\log p(\boldsymbol{x}_n;\boldsymbol{\theta}) \qquad \boldsymbol{\theta}_{MLE}=\operatorname*{argmin}_{\boldsymbol{\theta}} NLL(\boldsymbol{\theta})$$

最大似然估计

- $\mathcal{D}=\{(\boldsymbol{x}_n,y_n)\}_{n=1}^{N}$，似然函数有两种形式
  - 第一种

$$p(\mathcal{D};\boldsymbol{\theta})=\prod_{n=1}^{N}p(y_n|\boldsymbol{x}_n;\boldsymbol{\theta})$$

  - 第二种

$$p(\mathcal{D};\boldsymbol{\theta})=\prod_{n=1}^{N}p(\boldsymbol{x}_n,y_n;\boldsymbol{\theta})=\prod_{n=1}^{N}p(y_n|\boldsymbol{x}_n;\boldsymbol{\theta})\prod_{n=1}^{N}p(\boldsymbol{x}_n;\boldsymbol{\theta})$$

$$\propto\prod_{n=1}^{N}p(y_n|\boldsymbol{x}_n;\boldsymbol{\theta})$$

如果$\boldsymbol{\theta}$与$\boldsymbol{x}_n$的概率分布无关

- 为求解方便，可采用对数似然或负对数似然作为优化目标函数

最大似然估计的KL散度解读

- 最大似然估计使得估计出的概率分布能尽可能接近经验分布
- 经验分布：定义在数据集$\mathcal{D}=\{\boldsymbol{x}_n\}_{n=1}^N$上

$$\hat{p}_{\mathcal{D}}(\boldsymbol{x}) \triangleq \frac{1}{N}\sum_{n=1}^{N}\delta(\boldsymbol{x}-\boldsymbol{x}_n)$$

- 我们希望找到一个分布$q(\boldsymbol{x})=p(\boldsymbol{x};\boldsymbol{\theta})$，使得其与经验分布$\hat{p}_{\mathcal{D}}(\boldsymbol{x})$尽可能相似
  - 概率相似（差异）度量：KL散度

$$\mathrm{KL}(\hat{p}_{\mathcal{D}} \parallel q) \triangleq \sum_{\boldsymbol{x}} \hat{p}_{\mathcal{D}}(\boldsymbol{x}) \log \frac{\hat{p}_{\mathcal{D}}(\boldsymbol{x})}{q(\boldsymbol{x})} = \sum_{\boldsymbol{x}} \hat{p}_{\mathcal{D}}(\boldsymbol{x}) \log \hat{p}_{\mathcal{D}}(\boldsymbol{x}) - \sum_{\boldsymbol{x}} \hat{p}_{\mathcal{D}}(\boldsymbol{x}) \log q(\boldsymbol{x})$$

最大似然估计的KL散度解读

$$\mathrm{KL}(\hat{p}_{\mathcal{D}} \parallel q) \triangleq \sum_{\boldsymbol{x}} \hat{p}_{\mathcal{D}}(\boldsymbol{x}) \log \frac{\hat{p}_{\mathcal{D}}(\boldsymbol{x})}{q(\boldsymbol{x})} = \sum_{\boldsymbol{x}} \hat{p}_{\mathcal{D}}(\boldsymbol{x}) \log \hat{p}_{\mathcal{D}}(\boldsymbol{x}) - \sum_{\boldsymbol{x}} \hat{p}_{\mathcal{D}}(\boldsymbol{x}) \log q(\boldsymbol{x})$$

$$= -\mathbb{H}(\hat{p}_{\mathcal{D}}) - \frac{1}{N} \sum_{n=1}^{N} \log q(\boldsymbol{x}_n)$$

$$= -\mathbb{H}(\hat{p}_{\mathcal{D}}) - \frac{1}{N} \sum_{n=1}^{N} \log p(\boldsymbol{x}_n; \boldsymbol{\theta}) = \mathrm{const} + \frac{1}{N} NLL(\boldsymbol{\theta})$$

$$\min_{q} \mathrm{KL}(\hat{p}_{\mathcal{D}} \parallel q) = \min_{\boldsymbol{\theta}} NLL(\boldsymbol{\theta})$$

最大似然估计的KL散度解读

- 如果$\mathcal{D}=\{(\boldsymbol{x}_n,y_n)\}_{n=1}^N$，则定义经验分布如下

$$\hat{p}_{\mathcal{D}}(\boldsymbol{x},y)=\hat{p}_{\mathcal{D}}(y|\boldsymbol{x})\hat{p}_{\mathcal{D}}(\boldsymbol{x})\triangleq\frac{1}{N}\sum_{n=1}^{N}\delta(\boldsymbol{x}-\boldsymbol{x}_n)\,\delta(y-y_n)$$

$$\mathbb{E}_{\hat{p}_{\mathcal{D}}(\boldsymbol{x})}\left[\mathrm{KL}\left(\hat{p}_{\mathcal{D}}(y|\boldsymbol{x})\parallel q(y|\boldsymbol{x})\right)\right]\triangleq\sum_{\boldsymbol{x}}\hat{p}_{\mathcal{D}}(\boldsymbol{x})\left(\sum_{y}\hat{p}_{\mathcal{D}}(y|\boldsymbol{x})\log\frac{\hat{p}_{\mathcal{D}}(y|\boldsymbol{x})}{q(y|\boldsymbol{x})}\right)$$

$$q(y|\boldsymbol{x})\triangleq p(y|\boldsymbol{x};\boldsymbol{\theta})$$

$$=\sum_{\boldsymbol{x},y}\hat{p}_{\mathcal{D}}(\boldsymbol{x},y)\log\hat{p}_{\mathcal{D}}(y|\boldsymbol{x})-\sum_{\boldsymbol{x},y}\hat{p}_{\mathcal{D}}(\boldsymbol{x},y)\log q(y|\boldsymbol{x})$$

$$=\mathrm{const}-\frac{1}{N}\sum_{n=1}^{N}\log p(y_n|\boldsymbol{x}_n;\boldsymbol{\theta})=\mathrm{const}+\frac{1}{N}NLL(\boldsymbol{\theta})$$

ITML 最大似然估计：示例1

- 求解伯努利分布参数
  - 假设我们想知道一个硬币是否是均匀的。为此，我们通过一系列随机抛硬币的实验来进行推断
  - 令抛硬币正面朝上的概率为θ，正面朝下的概率为$1-\theta$。显然θ对我们而言是未知的，我们希望通过实验中观测到的结果来估计θ的值，进而判断该硬币是否均匀
  - 进一步，令$x \in \{0,1\}$是代表硬币正面朝向的随机变量， $x=1$代表正面朝上， $x=0$代表正面朝下。因此

$$p(x=1;\theta)=\theta$$

$$p(x=0;\theta)=1-\theta$$

$$x \sim \mathrm{Ber}(\theta)=\theta^{x}(1-\theta)^{1-x}$$

最大似然估计：示例1

- 假设我们抛了十次硬币，结果如下
  - $\{1, 1, 0, 0, 0, 1, 0, 1, 1, 1\}$
  - 即$\mathcal{D} = \{x_1 = 1, x_2 = 1, x_3 = 0, x_4 = 0, x_5 = 0, x_6 = 1, x_7 = 0, x_8 = 1, x_9 = 1, x_{10} = 1\}$

$$LL(\theta) = \log p(\mathcal{D};\theta) = \sum_{n=1}^{10} \log p(x_n;\theta)$$

$$= \sum_{x_n=1} \log p(x_n;\theta) + \sum_{x_n=0} \log p(x_n;\theta) = 6\log\theta + 4\log(1-\theta)$$

$$\frac{d}{d\theta} LL(\theta) = \frac{6}{\theta} - \frac{4}{1-\theta}$$

极值在目标函数导数或梯度上的值为零

注意：也可利用

$$\theta = \underset{\theta}{\operatorname{argmin}}\, NLL(\theta)$$

进行求解，方法基本一样

$$\theta = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, LL(\theta) \Rightarrow \frac{d}{d\theta} LL(\theta) = 0 \Rightarrow \frac{6}{\theta} - \frac{4}{1-\theta} = 0 \Rightarrow \theta = \frac{3}{5}$$

最大似然估计：示例1

- 推广到一般情形
  - $\mathcal{D} = \{x_n\}_{n=1}^{N}$， $x_n \in \{0,1\}$

$$LL(\theta) = \log p(\mathcal{D};\theta) = \sum_{n=1}^{N} \log p(x_n;\theta)$$

充分统计量
(sufficient statistics)

$$N_1 \triangleq |\{x_n | x_n = 1\}|$$

$$N_0 \triangleq |\{x_n | x_n = 0\}|$$

$$= \sum_{x_n=1} \log p(x_n;\theta) + \sum_{x_n=0} \log p(x_n;\theta) = N_1 \log\theta + N_0 \log(1-\theta)$$

$$\frac{d}{d\theta} LL(\theta) = \frac{N_1}{\theta} - \frac{N_0}{1-\theta}$$

$$\theta = \underset{\theta}{argmax}\, LL(\theta) \Longrightarrow \frac{d}{d\theta} NLL(\theta) = 0 \Longrightarrow \frac{N_1}{\theta} - \frac{N_0}{1-\theta} = 0 \Longrightarrow \theta = \frac{N_1}{N}$$

ITML 最大似然估计：示例2

- 求解类型分布参数
  - 假设我们想知道一个K面的骰子是否是均匀的。为此，我们通过一系列随机掷骰子的实验来进行推断
  - 令掷骰子$k(1 \le k \le K)$面朝上（出现）的概率为θ_k，我们希望通过实验观测到的结果来估计θ_k的值，进而判断该骰子是否均匀

$$p(x) = \mathrm{Cat}(x; \boldsymbol{\theta}) = \prod_{k=1}^{K} \theta_k^{\mathbb{I}(x=k)}$$

  - 进一步，我们随机掷骰子N次，得到结果数据集$\mathcal{D} = \{x_n\}_{n=1}^{N}$， $x_n \in \{1, \dots, K\}$
  - 故（注意：这里我们采用负对数似然来实现最大似然估计， $\boldsymbol{\theta} = \{\theta_k\}$）

$$NLL(\boldsymbol{\theta}) = -\log p(\mathcal{D}; \boldsymbol{\theta}) = -\sum_{n=1}^{N} \log p(x_n; \boldsymbol{\theta})$$

最大似然估计：示例2

$$NLL(\boldsymbol{\theta}) = -\log p(\mathcal{D};\theta) = -\sum_{n=1}^{N}\log p(x_n;\theta)$$

$$= -\sum_{n=1}^{N}\log\prod_{k=1}^{K}\theta_k^{\mathbb{I}(x_n=k)} = -\sum_{k=1}^{K}\sum_{n=1}^{N}\mathbb{I}(x_n=k)\log\theta_k = -\sum_{k=1}^{K}N_k\log\theta_k$$

充分统计量

$$N_k \triangleq \sum_{n=1}^{N}\mathbb{I}(x_n=k)$$

$$\boldsymbol{\theta}_{MLE} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\, NLL(\boldsymbol{\theta})$$

带约束条件： $\sum_{k=1}^{K}\theta_k = 1$

拉格朗日乘子法

$$\mathcal{L}(\boldsymbol{\theta},\lambda) = NLL(\boldsymbol{\theta}) + \lambda\left(1-\sum_{k=1}^{K}\theta_k\right)$$

求解

$$\begin{cases}\dfrac{\partial}{\partial\theta_i}\mathcal{L}(\boldsymbol{\theta},\lambda) = 0\\[2ex]\dfrac{\partial}{\partial\lambda}\mathcal{L}(\boldsymbol{\theta},\lambda) = 0\end{cases}$$

最大似然估计：示例2

$$\frac{\partial}{\partial \theta_i}\mathcal{L}(\boldsymbol{\theta},\lambda) = \frac{\partial}{\partial \theta_i} - \sum_{k=1}^{K} N_k \log \theta_k + \frac{\partial}{\partial \theta_i}\lambda\left(\sum_{k=1}^{K}\theta_k - 1\right) = -\frac{N_i}{\theta_i} + \lambda$$

$$\frac{\partial}{\partial \lambda}\mathcal{L}(\boldsymbol{\theta},\lambda) = 1 - \sum_{k=1}^{K}\theta_k$$

$$\left.\begin{aligned}\frac{\partial}{\partial \theta_i}\mathcal{L}(\boldsymbol{\theta},\lambda) &= 0\\ \frac{\partial}{\partial \lambda}\mathcal{L}(\boldsymbol{\theta},\lambda) &= 0\end{aligned}\right\} \Rightarrow \left.\begin{aligned}\boldsymbol{\theta}_i &= \frac{N_i}{\lambda}\\ \sum_{k=1}^{K}\boldsymbol{\theta}_i &= 1\end{aligned}\right\} \Rightarrow \boldsymbol{\theta}_i = \frac{N_i}{\sum_{k=1}^{K} N_k} = \frac{N_i}{N}$$

最大似然估计：示例3

- 求解高斯分布参数
  - 假设$x \sim \mathcal{N}(\mu, \sigma^2)$，$\mu$ 和σ^2未知。
  - 虽然均值和方差不知道，但从这个分布中随机采样了N个点。这些点记为$\mathcal{D} = \{x_n\}_{n=1}^N$。我们希望用数据集$\mathcal{D}$来估计$\mu$ 和σ^2的值
  - 故（$\boldsymbol{\theta} = \{\mu, \sigma^2\}$）

$$NLL(\boldsymbol{\theta}) = -\log p(\mathcal{D}; \boldsymbol{\theta}) = -\sum_{n=1}^{N} \log p(x_n; \boldsymbol{\theta})$$

$$= \frac{N}{2}\log 2\pi + \frac{N}{2}\log\sigma^2 + \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n - \mu)^2$$

最大似然估计：示例3

$$\boldsymbol{\theta}_{MLE} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} NLL(\boldsymbol{\theta}) \Longrightarrow \text{求解} \begin{cases} \dfrac{\partial}{\partial \mu} NLL(\boldsymbol{\theta}) = 0 \\ \dfrac{\partial}{\partial \sigma^2} NLL(\boldsymbol{\theta}) = 0 \end{cases}$$

$$NLL(\boldsymbol{\theta}) = \frac{N}{2}\log 2\pi + \frac{N}{2}\log\sigma^2 + \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n - \mu)^2$$

$$\frac{\partial}{\partial \mu} NLL(\boldsymbol{\theta}) = 0 \Longrightarrow \mu = \boxed{\frac{1}{N}\sum_{n=1}^{N} x_n}$$

充分统计量

$$\frac{\partial}{\partial \sigma^2} NLL(\boldsymbol{\theta}) = 0 \Longrightarrow \sigma^2 = \frac{1}{N}\sum_{n=1}^{N}(x_n - \mu)^2 = \boxed{\frac{1}{N}\sum_{n=1}^{N} {x_n}^2} - \left(\frac{1}{N}\sum_{n=1}^{N} x_n\right)^2$$

最大似然估计：问题

- 最大似然估计的优化目标是尽量使得看到的数据出现的概率最大。这导致其对未见到的数据难于给出合适得概率估计，造成过拟合现象（overfitting，或称过学习）
- 例子：
  - 假设我们抛了五次硬币，结果五次都正面朝上，即$\mathcal{D} = \{1, 1, 1, 1, 1\}$
  - 按照最大似然估计，可知硬币正面朝上的概率为1，朝下的概率为0
  - 这意味着，如果我们根据最大似然估计的结果进行预测，无论抛多少次硬币，我们都将预测正面朝上。显然，这与我们的实际感受和直觉不一致（小概率事件也是有可能发生的）
- 问题关键
  - 最大似然估计的优化目标仅限于拟合经验分布，而经验分布并非数据的真实分布
  - 解决之道：引入先验知识

内容

- 最大似然估计
  - 基本概念
  - KL散度解读
  - 示例
- 最大后验估计
  - 基本概念
  - 示例
- 贝叶斯方法
  - 基本概念
  - 共轭先验

最大后验估计

- Maximum a posterior estimation (MAP)
- 把待估计的参数$\boldsymbol{\theta}$视为随机变量，且服从某个先验分布
  - 参数的先验分布可视为我们在未看到数据集$\mathcal{D}$之前对参数的先验知识
- 看到数据集$\mathcal{D}$之后，我们对参数的分布有了新的认识，即其后验分布。由贝叶斯公式可知

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}$$

  - 显然，根据$p(\boldsymbol{\theta}|\mathcal{D})$对参数$\boldsymbol{\theta}$进行采样，最有可能选择的样本点是$p(\boldsymbol{\theta}|\mathcal{D})$的众数（最大值），以$p(\boldsymbol{\theta}|\mathcal{D})$的众数来配置采样出数据集$\mathcal{D}$的概率分布的参数，称为最大后验估计

$$\boldsymbol{\theta}_{MAP} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, p(\boldsymbol{\theta}|\mathcal{D}) = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})$$

最大后验估计

$$\boldsymbol{\theta}_{MAP}=\underset{\boldsymbol{\theta}}{\operatorname{argmax}}\,p(\boldsymbol{\theta}|\mathcal{D})=\underset{\boldsymbol{\theta}}{\operatorname{argmax}}\frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}=\underset{\boldsymbol{\theta}}{\operatorname{argmax}}\,p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})$$

$$\max_{\boldsymbol{\theta}} p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta}) \Leftrightarrow \max_{\boldsymbol{\theta}}\left(\log p(\mathcal{D}|\boldsymbol{\theta})+\boxed{\log p(\boldsymbol{\theta})}\right)$$

正则项，后续课程有详述

$$\boldsymbol{\theta}_{MAP}=\underset{\boldsymbol{\theta}}{\operatorname{argmax}}\left(\log p(\mathcal{D}|\boldsymbol{\theta})+\log p(\boldsymbol{\theta})\right)$$

注意：给定$\boldsymbol{\theta}$，$p(\mathcal{D}|\boldsymbol{\theta})$就是$p(\mathcal{D};\boldsymbol{\theta})$

$$\boldsymbol{\theta}_{MAP}=\underset{\boldsymbol{\theta}}{\operatorname{argmax}}\left(LL(\boldsymbol{\theta})+\log p(\boldsymbol{\theta})\right)=\underset{\boldsymbol{\theta}}{\operatorname{argmax}}\log\left(p(\mathcal{D};\boldsymbol{\theta})+\log p(\boldsymbol{\theta})\right)$$

当$\boldsymbol{\theta}$的先验分布$p(\boldsymbol{\theta})$是均匀分布时， $\boldsymbol{\theta}_{MAP}=\boldsymbol{\theta}_{MLE}$

ITML 最大后验估计：示例1

- 求解硬币正面朝上的概率
  - 令抛硬币正面朝上的概率为θ。显然θ对我们而言是未知的，我们希望通过观测抛硬币试验中观测到的结果来估计θ的值，进而判断该硬币是否均匀
  - 进一步，令$x \in \{0,1\}$是代表硬币正面朝向的随机变量， $x=1$代表正面朝上， $x=0$代表正面朝下。正面朝上的概率为θ。即$p(x=1;\theta)=\theta$
  - 假定$\theta \sim \mathrm{Beta}(a,b)(a>0, b>0)$

$$p(\theta)=\mathrm{Beta}(\theta;a,b)=\frac{1}{\mathrm{B}(a,b)}\theta^{a-1}(1-\theta)^{b-1}$$

$$\mathrm{B}(a,b)=\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

$$\Gamma(a)=\int_0^{\infty} x^{a-1}e^{-x}dx$$

  - 假定抛了十次硬币，结果为$\mathcal{D}=\{1,1,0,0,0,1,0,1,1,1\}$

最大后验估计：示例1

$$\mathcal{D} = \{1,1,0,0,0,1,0,1,1,1\}$$

$$L_{MAP}(\theta) \triangleq \log p(\mathcal{D}|\boldsymbol{\theta}) + \log p(\boldsymbol{\theta})$$

$$= \sum_{x_n=1} \log p(x_n;\theta) + \sum_{x_n=0} \log p(x_n;\theta) - \log \mathrm{B}(a,b) + (a-1)\log\theta + (b-1)\log(1-\theta)$$

$$= (a+5)\log\theta + (b+3)\log(1-\theta) - \log \mathrm{B}(a,b)$$

$$\frac{d}{d\theta} L_{MAP}(\theta) = \frac{a+5}{\theta} - \frac{b+3}{1-\theta}$$

$$\theta_{MAP} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, L_{MAP}(\theta) \Rightarrow \frac{d}{d\theta} L_{MAP}(\theta) = 0 \Rightarrow \frac{a+5}{\theta} - \frac{b+3}{1-\theta} = 0 \Rightarrow \theta = \frac{a+5}{a+b+8}$$

ITML 最大后验估计：示例1

- 推广到一般情形
  - $\mathcal{D}=\{x_n\}_{n=1}^{N}$， $x_n\in\{0,1\}$

$$N_1 \triangleq |\{x_n|x_n=1\}|$$

$$N_0 \triangleq |\{x_n|x_n=0\}|$$

$$L_{MAP}(\theta)\triangleq \log p(\mathcal{D}|\boldsymbol{\theta})+\log p(\boldsymbol{\theta})$$

$$=\sum_{x_n=1}\log p(x_n;\theta)+\sum_{x_n=0}\log p(x_n;\theta)-\log \mathrm{B}(a,b)+(a-1)\log\theta+(b-1)\log(1-\theta)$$

$$=(N_1+a-1)\log\theta+(N_0+b-1)\log(1-\theta)-\log \mathrm{B}(a,b)$$

$$\frac{d}{d\theta}L_{MAP}(\theta)=\frac{N_1+a-1}{\theta}-\frac{N_0+b-1}{1-\theta}$$

加一平滑

add-one smoothing

$$\theta=\frac{N_1+1}{N+2}$$

$$\Uparrow \quad a=b=2$$

$$\theta=\underset{\theta}{argmax}\,L_{MAP}(\theta)\Longrightarrow\frac{d}{d\theta}L_{MAP}(\theta)=0\Longrightarrow\frac{N_1+a-1}{\theta}-\frac{N_0+b-1}{1-\theta}=0\Longrightarrow\theta=\frac{N_1+a-1}{N+a+b-2}$$

最大后验估计：讨论

- 加一平滑
  - 一类简单但应用广泛的技术，可用来有效避免零计数问题 (zero count problem)
- 零计数问题
  - 某些数据或信息没有在训练数据集中出现，故没有统计信息。这在更广泛的层面上讲，是学习的泛化问题或过学习问题。这个问题类似于哲学中的“黑天鹅悖论”(black swan paradox)
- 黑天鹅悖论
  - 古代西方人看到的所有天鹅都是白色的。因此，认为黑天鹅是不存在的。借黑天鹅指代不可能发生的事情
  - 这个悖论展示了归纳学习的根本性问题：从过去观察到的现象（数据）去归纳出关于未来的一般性结论
    - ✓这在逻辑上是有瑕疵的
    - ✓目前，机器学习主体是基于归纳学习的。为克服这个问题，最好在学习过程中融合经验数据和先验知识。

最大后验估计：示例2

- 求解高斯分布参数
  - 假设$x \sim \mathcal{N}(\mu, \sigma^2)$，$\mu$未知，$\sigma^2$是一已知常量，$\boldsymbol{\theta} = \mu$
  - 假定$\mu \sim \mathcal{N}(\mu_1, \sigma_1^2)$
  - 采样数据集$\mathcal{D} = \{x_n\}_{n=1}^N$
  - 通过最大后验估计μ

$$L_{MAP}(\mu) \triangleq \log p(\mathcal{D}|\mu, \sigma^2)p(\mu) = \log p(\mathcal{D}|\mu, \sigma^2) + \log p(\mu)$$

$$= \sum_{n=1}^{N} \log \mathcal{N}(x_n; \mu, \sigma^2) + \log \mathcal{N}(\mu; \mu_1, \sigma_1^2)$$

最大似然估计：示例2

$$L_{MAP}(\mu, \sigma^2) = \sum_{n=1}^{N} \log \mathcal{N}(x_n; \mu, \sigma^2) + \log \mathcal{N}(\mu; \mu_1, \sigma_1^2)$$

$$= -\frac{N}{2}\log 2\pi\sigma^2 - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n - \mu)^2 - \frac{1}{2}\log 2\pi\sigma_1^2 - \frac{1}{2\sigma_1^2}(\mu - \mu_1)^2$$

$$0 = \frac{d}{d\mu}L_{MAP}(\mu, \sigma^2) = \frac{1}{\sigma^2}\sum_{n=1}^{N}(x_n - \mu) - \frac{1}{\sigma_1^2}(\mu - \mu_1)$$

$$\Downarrow$$

$$\mu = \frac{\frac{1}{\sigma^2}\sum_{n=1}^{N} x_n + \frac{1}{\sigma_1^2}\mu_1}{\frac{N}{\sigma^2} + \frac{1}{\sigma_1^2}}$$

内容

- 最大似然估计
  - 基本概念
  - KL散度解读
  - 示例
- 最大后验估计
  - 基本概念
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- 贝叶斯方法
  - 基本概念
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贝叶斯方法

- 之前介绍了从数据中估计概率模型参数的基本方法。但这些方法忽略了估计量的不确定性。对不确定性的把握，在一些应用中非常重要
  - 自动驾驶、医疗健康、经济金融等
- 贝叶斯方法
  - 也称贝叶斯统计方法，是处理统计量不确定性的一类方法
  - 把统计量（模型参数）视为随机变量，用模型参数的后验概率分布来表达和处理不确定性
  - 基本框架：为计算后验，先假定模型参数$\boldsymbol{\theta}$服从某个先验分布$p(\boldsymbol{\theta})$，表示在未看到数据之前，我们对参数的先验认知。然后，定义似然函数$p(\mathcal{D}|\boldsymbol{\theta})$，反映在当前参数配置下，数据集$\mathcal{D}$生成的概率。最后通过贝叶斯公式，计算后验概率

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})} \qquad p(\mathcal{D}) = \sum_{\boldsymbol{\theta}^*} p(\mathcal{D},\boldsymbol{\theta}^*) \qquad p(\mathcal{D}) = \int p(\mathcal{D},\boldsymbol{\theta}^*)\, d\boldsymbol{\theta}^*$$

✓注意：最大后验估计仅需求解出最大后验值即可，不需精确得出后验分布

贝叶斯方法

- 求出后验分布$p(\boldsymbol{\theta}|\mathcal{D})$后，对给定的输入数据$\boldsymbol{x}$，我们可以如下计算其后验预测分布
  - 无监督场景：$\mathcal{D}=\{\boldsymbol{x}_n\}_{n=1}^N$

$$\begin{aligned} p(\boldsymbol{x}|\mathcal{D}) &= \int p(\boldsymbol{x},\boldsymbol{\theta}|\mathcal{D})\,d\boldsymbol{\theta} \\ &= \int p(\boldsymbol{x}|\boldsymbol{\theta},\mathcal{D})p(\boldsymbol{\theta}|\mathcal{D})\,d\boldsymbol{\theta} \\ &= \int p(\boldsymbol{x}|\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathcal{D})\,d\boldsymbol{\theta} \\ &= \mathbb{E}_{\boldsymbol{\theta}\sim p(\boldsymbol{\theta}|\mathcal{D})}[p(\boldsymbol{x}|\boldsymbol{\theta})] \end{aligned}$$

$$\begin{aligned} p(\boldsymbol{x}|\mathcal{D}) &= \sum_{\boldsymbol{\theta}} p(\boldsymbol{x},\boldsymbol{\theta}|\mathcal{D}) \\ &= \sum_{\boldsymbol{\theta}} p(\boldsymbol{x}|\boldsymbol{\theta},\mathcal{D})p(\boldsymbol{\theta}|\mathcal{D}) \\ &= \sum_{\boldsymbol{\theta}} p(\boldsymbol{x}|\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathcal{D}) \\ &= \mathbb{E}_{\boldsymbol{\theta}\sim p(\boldsymbol{\theta}|\mathcal{D})}[p(\boldsymbol{x}|\boldsymbol{\theta})] \end{aligned}$$

  - 通过以上形式来进行预测的机器学习方法也称为贝叶斯机器学习 (Bayesian machine learning)
  - 上述方法实际上是用模型预测结果的期望（加权平均）来进行最终预测，而不是仅仅用一个“最好”的参数模型（如MLE或MAP）来进行预测。因此，贝叶斯机器学习方法能有效规避过学习问题

- 对于监督场景下，即$\mathcal{D}=\{(\boldsymbol{x}_n,y_n)\}_{n=1}^N$，在求出后验分布$p(\boldsymbol{\theta}|\mathcal{D})$后，对给定的输入数据$\boldsymbol{x}$，我们可以如下计算其标签后验预测分布

$$
\begin{aligned}
p(y|\boldsymbol{x},\mathcal{D}) &= \int p(y,\boldsymbol{\theta}|\boldsymbol{x},\mathcal{D})\,d\boldsymbol{\theta} \\
&= \int p(y|\boldsymbol{x},\boldsymbol{\theta},\mathcal{D})p(\boldsymbol{\theta}|\boldsymbol{x},\mathcal{D})\,d\boldsymbol{\theta} \\
&= \int p(y|\boldsymbol{x},\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathcal{D})\,d\boldsymbol{\theta} \\
&= \mathbb{E}_{\boldsymbol{\theta}\sim p(\boldsymbol{\theta}|\mathcal{D})}[p(y|\boldsymbol{x},\boldsymbol{\theta})]
\end{aligned}
$$

$$
\begin{aligned}
p(y|\boldsymbol{x},\mathcal{D}) &= \sum_{\boldsymbol{\theta}} p(y,\boldsymbol{\theta}|\boldsymbol{x},\mathcal{D}) \\
&= \sum_{\boldsymbol{\theta}} p(y|\boldsymbol{x},\boldsymbol{\theta},\mathcal{D})p(\boldsymbol{\theta}|\boldsymbol{x},\mathcal{D}) \\
&= \sum_{\boldsymbol{\theta}} p(y|\boldsymbol{x},\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathcal{D}) \\
&= \mathbb{E}_{\boldsymbol{\theta}\sim p(\boldsymbol{\theta}|\mathcal{D})}[p(y|\boldsymbol{x},\boldsymbol{\theta})]
\end{aligned}
$$

贝叶斯方法

- 核心问题
  - 给定似然函数$p(\mathcal{D}|\boldsymbol{\theta})$和先验分布$p(\boldsymbol{\theta})$。理论上我们可由贝叶斯公式来计算后验概率$p(\boldsymbol{\theta}|\mathcal{D})$
  - 但是，上述过程需要计算$p(\mathcal{D})$，而$p(\mathcal{D})$是一个求和或求积分的过程。求解$p(\mathcal{D})$在实际应用中的计算代价非常高，甚至是不可行的
- $p(\boldsymbol{\theta}|\mathcal{D})$的两类计算方法
  - 精确计算方法
    - ✓共轭先验方法 (conjugate priors)
  - 近似计算方法
    - ✓变分近似 (Variational approximation)
    - ✓MCMC近似 (Markov Chain Monte Carlo approximation)

ITML 共轭先验

- 假定先验分布$p(\boldsymbol{\theta}) \in \mathcal{F}$（$\mathcal{F}$是一分布族）。称$p(\boldsymbol{\theta})$是似然函数$p(\mathcal{D}|\boldsymbol{\theta})$的共轭先验，如果后验分布$p(\boldsymbol{\theta}|\mathcal{D})$也属于$\mathcal{F}$，即$p(\boldsymbol{\theta}|\mathcal{D}) \in \mathcal{F}$
  - 即，先验分布和后验分布属于同一个分布族
  - 同时，也意味着$\mathcal{F}$在贝叶斯公式下是封闭的
  - 设计好先验分布和似然函数，形成共轭关系，则后验分布可直接求解
- 如果$\mathcal{F}$是指数分布族，则后验分布有闭式解（可直接求解）
- 三个共轭模型
  - Beta-Bernoulli模型
  - Dirichlet-categorical模型
  - Gaussian-Gaussian模型

Beta-Bernoulli模型

- 由于伯努利分布是简单的二项分布，故Beta-Bernoulli模型结果很容易推广到Beta-binomial模型
- 假定抛硬币N次，得到观察结果$\mathcal{D}=\{x_n\}_{n=1}^N$。其中$x_n$表示第$n$次的结果。$x_n=1$代表正面朝上，$x_n=0$代表正面朝下。令正面朝上的概率为$\theta$，则$x_n \sim \mathrm{Ber}(\theta)$
- 伯努利似然函数

$$p(\mathcal{D}|\theta)=\prod_{n=1}^{N}\theta^{x_n}(1-\theta)^{1-x_n}=\theta^{N_1}(1-\theta)^{N_0}$$

$$N_1 \triangleq |\{x_n|x_n=1\}|$$

$$N_0 \triangleq |\{x_n|x_n=0\}|$$

- 先验分布
  - 为了保证与伯努利似然函数共轭，θ的先验分布应该具有如下的形式

$$p(\theta) \propto \theta^u(1-\theta)^v$$

  - Beta分布具备上述形式。记$\theta \sim \mathrm{Beta}(\alpha,\beta)$

$$p(\theta)=\mathrm{Beta}(\theta;\alpha,\beta)=\frac{1}{\mathrm{B}(\alpha,\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}$$

- 后验分布

$$\mathrm{Beta}(\theta;\alpha,\beta) \triangleq \frac{1}{\mathrm{B}(\alpha,\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}$$

$$\begin{aligned}
p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})} &\propto p(\mathcal{D}|\theta)p(\theta) \\
&\propto \theta^{N_1}(1-\theta)^{N_0}\theta^{\alpha-1}(1-\theta)^{\beta-1} \\
&= \theta^{N_1+\alpha-1}(1-\theta)^{N_0+\beta-1} \\
&\propto \mathrm{Beta}(\theta;N_1+\alpha,N_0+\beta)
\end{aligned}$$

先验分布中的参数α和β称为超参数

置$\alpha=\beta=2$，表达先验倾向θ取1/2

置$\alpha=\beta=1$。先验分布是均匀分布，表达先验对θ取值无倾向

$$\left.\begin{aligned}
p(\theta|\mathcal{D}) &= C\times\mathrm{Beta}(\theta;N_1+\alpha,N_0+\beta) \\
\int p(\theta|\mathcal{D})d\theta &= 1 \\
\int \mathrm{Beta}(\theta;N_1+\alpha,N_0+\beta)d\theta &= 1
\end{aligned}\right\} \Rightarrow p(\theta|\mathcal{D}) = \mathrm{Beta}(\theta;N_1+\alpha,N_0+\beta)$$

ITML 后验分布统计量：众数

- 后验分布众数 (posterior mode)

$$\alpha = \beta = 2$$

$$\alpha = \beta = 1$$

$$\theta_{mode}?$$

$$p(\theta|\mathcal{D}) = \mathrm{Beta}(\theta; N_1 + \alpha, N_0 + \beta)$$

$$= \frac{1}{\mathrm{B}(N_1 + \alpha, N_0 + \beta)} \theta^{N_1+\alpha-1}(1-\theta)^{N_0+\beta-1}$$

θ_{MAP}

$$\theta_{mode} = \max_{\theta} p(\theta|\mathcal{D}) = \max_{\theta} \mathrm{Beta}(\theta; N_1 + \alpha, N_0 + \beta)$$

$$\frac{d}{d\theta}\mathrm{Beta}(\theta; N_1 + \alpha, N_0 + \beta) = \theta^{N_1+\alpha-1}(1-\theta)^{N_0+\beta-1}\left(\frac{N_1+\alpha-1}{\theta} - \frac{N_0+\beta-1}{1-\theta}\right)$$

$$\theta_{mode} = \underset{\theta}{argmax}\, p(\theta|\mathcal{D}) \Longrightarrow \frac{d}{d\theta}\mathrm{Beta}(\theta_{mode}; N_1 + \alpha, N_0 + \beta) = 0 \Longrightarrow \frac{N_1+\alpha-1}{\theta_{mode}} - \frac{N_0+\beta-1}{1-\theta_{mode}} = 0$$

$$\Longrightarrow \theta_{mode} = \frac{N_1+\alpha-1}{N_1+\alpha+N_0+\beta-2} = \frac{N_1+\alpha-1}{N+\alpha+\beta-2}$$

ITML 后验分布统计量：均值

- 后验分布均值
  - 相比众数，均值是更为鲁棒的统计量

$$\bar{\theta} \triangleq \mathbb{E}[\theta|\mathcal{D}] \qquad p(\theta|\mathcal{D}) = \mathrm{Beta}(\theta; N_1+\alpha, N_0+\beta) = \frac{1}{\mathrm{B}(N_1+\alpha, N_0+\beta)}\theta^{N_1+\alpha-1}(1-\theta)^{N_0+\beta-1}$$

记$p(\theta) \triangleq \mathrm{Beta}(\theta;\ \alpha,\beta) = \frac{1}{\mathrm{B}(\alpha,\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}$

$$\mathbb{E}[\theta] = \int \frac{1}{\mathrm{B}(\alpha,\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}\theta d\theta = \frac{1}{\mathrm{B}(\alpha,\beta)}\int \theta^{(\alpha+1)-1}(1-\theta)^{\beta-1}d\theta$$

$$\mathrm{Beta}(\theta;\ \alpha+1,\beta) = \frac{1}{\mathrm{B}(\alpha+1,\beta)}\theta^{(\alpha+1)-1}(1-\theta)^{\beta-1}$$

ITML 后验分布统计量：均值

$$1 = \int \frac{1}{\mathrm{B}(\alpha+1,\beta)} \theta^{(\alpha+1)-1}(1-\theta)^{\beta-1} d\theta \Longrightarrow \mathrm{B}(\alpha+1,\beta) = \int \theta^{(\alpha+1)-1}(1-\theta)^{\beta-1} d\theta$$

$$\mathbb{E}[\theta] = \frac{1}{\mathrm{B}(\alpha,\beta)} \int \theta^{(\alpha+1)-1}(1-\theta)^{\beta-1} d\theta = \frac{\mathrm{B}(\alpha+1,\beta)}{\mathrm{B}(\alpha,\beta)}$$

$$\mathrm{B}(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

$$\mathbb{E}[\theta] = \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+1+\beta)} \times \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} = \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)} \times \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha+\beta+1)}$$

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx \Longrightarrow \Gamma(\alpha+1) = \alpha\Gamma(\alpha)$$

$$\Rightarrow \mathbb{E}[\theta] = \frac{\alpha}{\alpha+\beta}$$

$$p(\theta|\mathcal{D}) = \mathrm{Beta}(\theta; N_1+\alpha, N_0+\beta) \Rightarrow \bar{\theta} \triangleq \mathbb{E}[\theta|\mathcal{D}] = \frac{N_1+\alpha}{N_1+\alpha+N_0+\beta} = \frac{N_1+\alpha}{N+\alpha+\beta}$$

后验分布统计量：均值

$$\bar{\theta} \triangleq \mathbb{E}[\theta|\mathcal{D}] = \frac{N_1+\alpha}{N_1+\alpha+N_0+\beta} = \frac{N_1+\alpha}{N+\alpha+\beta}$$

$$\theta_{MLE} = \frac{N_1}{N}$$

记 $\tilde{\theta} \triangleq \frac{\alpha}{\alpha+\beta}$，　$\tilde{N} = \alpha + \beta$

$$\bar{\theta} = \frac{N_1+\alpha}{N+\alpha+\beta} = \frac{N_1+\tilde{\theta}\tilde{N}}{N+\tilde{N}} = \frac{\tilde{N}}{N+\tilde{N}}\tilde{\theta} + \frac{N}{N+\tilde{N}}\frac{N_1}{N}$$

$$\lambda \triangleq \frac{\tilde{N}}{N+\tilde{N}}$$

$$= \lambda\tilde{\theta} + (1-\lambda)\theta_{MLE}$$

先验均值

后验分布统计量：方差

- 后验分布方差 $p(\theta|\mathcal{D}) = \mathrm{Beta}(\theta; N_1+\alpha, N_0+\beta) = \frac{1}{\mathrm{B}(N_1+\alpha, N_0+\beta)}\theta^{N_1+\alpha-1}(1-\theta)^{N_0+\beta-1}$
  - 刻画估值的不确定性

$$\mathbb{V}[\theta|\mathcal{D}] = \mathbb{E}[\theta^2|\mathcal{D}] - (\mathbb{E}[\theta|\mathcal{D}])^2$$

$$\mathbb{E}[\theta|\mathcal{D}] = \frac{N_1+\alpha}{N+\alpha+\beta}$$

$$\mathbb{E}[\theta^2|\mathcal{D}] = \int \frac{1}{\mathrm{B}(N_1+\alpha, N_0+\beta)}\theta^{N_1+\alpha-1}(1-\theta)^{N_0+\beta-1}\theta^2 d\theta$$

$$= \frac{1}{\mathrm{B}(N_1+\alpha, N_0+\beta)}\int \theta^{N_1+\alpha+2-1}(1-\theta)^{N_0+\beta-1} d\theta$$

$$= \frac{\mathrm{B}(N_1+\alpha+2, N_0+\beta)}{\mathrm{B}(N_1+\alpha, N_0+\beta)} = \frac{\Gamma(N_1+\alpha+2)\Gamma(N_0+\beta)}{\Gamma(N_1+N_0+\alpha+\beta+2)} \times \frac{\Gamma(N_1+N_0+\alpha+\beta)}{\Gamma(N_1+\alpha)\Gamma(N_0+\beta)}$$

后验分布统计量：方差

$$\mathbb{E}[\theta^2|\mathcal{D}] = \frac{\Gamma(N_1+\alpha+2)\Gamma(N_0+\beta)}{\Gamma(N_1+N_0+\alpha+\beta+2)} \times \frac{\Gamma(N_1+N_0+\alpha+\beta)}{\Gamma(N_1+\alpha)\Gamma(N_0+\beta)} = \frac{\Gamma(N_1+\alpha+2)}{\Gamma(N_1+\alpha)} \times \frac{\Gamma(N_1+N_0+\alpha+\beta)}{\Gamma(N_1+N_0+\alpha+\beta+2)}$$

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx \Rightarrow \Gamma(\alpha+1) = \alpha\Gamma(\alpha) \Rightarrow \Gamma(\alpha+2) = (\alpha+1)\Gamma(\alpha+1) = (\alpha+1)\alpha\Gamma(\alpha)$$

$$\mathbb{E}[\theta^2|\mathcal{D}] = \frac{(N_1+\alpha+1)(N_1+\alpha)}{(N_1+N_0+\alpha+\beta+1)(N_1+N_0+\alpha+\beta)} = \frac{(N_1+\alpha+1)(N_1+\alpha)}{(N+\alpha+\beta+1)(N+\alpha+\beta)}$$

$$\mathbb{V}[\theta|\mathcal{D}\,] = \mathbb{E}[\theta^2|\mathcal{D}] - (\mathbb{E}[\theta|\mathcal{D}])^2 = \frac{(N_1+\alpha+1)(N_1+\alpha)}{(N+\alpha+\beta+1)(N+\alpha+\beta)} - \left(\frac{N_1+\alpha}{N+\alpha+\beta}\right)^2$$

$$= \frac{N_1+\alpha}{N+\alpha+\beta}\left(\frac{N_1+\alpha+1}{N+\alpha+\beta+1} - \frac{N_1+\alpha}{N+\alpha+\beta}\right)$$

后验分布统计量：方差

$$\mathbb{V}[\theta|\mathcal{D}] = \frac{N_1+\alpha}{N+\alpha+\beta}\left(\frac{N_1+\alpha+1}{N+\alpha+\beta+1} - \frac{N_1+\alpha}{N+\alpha+\beta}\right)$$

记 $\hat{\alpha} \triangleq N_1+\alpha, \hat{\beta} \triangleq N_0+\beta$

$$\mathbb{E}[\theta|\mathcal{D}] = \frac{N_1+\alpha}{N+\alpha+\beta} = \frac{\hat{\alpha}}{\hat{\alpha}+\hat{\beta}} \qquad \mathbb{V}[\theta|\mathcal{D}] = \frac{\hat{\alpha}}{\hat{\alpha}+\hat{\beta}}\left(\frac{\hat{\alpha}+1}{\hat{\alpha}+\hat{\beta}+1} - \frac{\hat{\alpha}}{\hat{\alpha}+\hat{\beta}}\right)$$

$$= \left(\frac{\hat{\alpha}}{\hat{\alpha}+\hat{\beta}}\right)^2 \frac{\hat{\beta}}{\hat{\alpha}(\hat{\alpha}+\hat{\beta}+1)} = \frac{\hat{\beta}}{\hat{\alpha}(\hat{\alpha}+\hat{\beta}+1)}(\mathbb{E}[\theta|\mathcal{D}])^2$$

如果 $N \gg \alpha+\beta$(或$N_1 \gg \alpha, N_0 \gg \beta$)，则

$$\mathbb{E}[\theta|\mathcal{D}] \approx \frac{N_1}{N} \qquad \frac{\hat{\beta}}{\hat{\alpha}(\hat{\alpha}+\hat{\beta}+1)} \approx \frac{N_0}{N_1 N} \qquad \mathbb{V}[\theta|\mathcal{D}] \approx \frac{N_0 N_1}{N^3} = \frac{\theta_{MLE}(1-\theta_{MLE})}{N}$$

标准差$\mathrm{std}[\theta|\mathcal{D}] = \sqrt{\mathbb{V}[\theta|\mathcal{D}]} = \sqrt{\frac{\theta_{MLE}(1-\theta_{MLE})}{N}}$，不确定性以$1/\sqrt{N}$的速度下降

后验预测

$$
\begin{aligned}
p(x=1|\mathcal{D}) &= \int p(x=1,\theta|\mathcal{D})\,d\theta \\
&= \int p(x=1|\theta,\mathcal{D})p(\theta|\mathcal{D})\,d\theta \\
&= \int p(x=1|\theta)p(\theta|\mathcal{D})\,d\theta \\
&= \int \theta \mathrm{Beta}(\theta; N_1+\alpha, N_0+\beta)\,d\theta \\
&= \mathbb{E}[\theta|\mathcal{D}] = \frac{N_1+\alpha}{N+\alpha+\beta} = \frac{\hat{\alpha}}{\hat{\alpha}+\hat{\beta}}
\end{aligned}
$$

如果$\theta\sim\mathrm{Beta}(1,1)$，即参数先验服从均匀分布，则$p(x=1|\mathcal{D})=\frac{N_1+1}{N+2}$。这就是之前最大后验估计时提到的加一平滑

注意：在最大后验估计时，我们是通过假定$\theta\sim\mathrm{Beta}(2,2)$来获得加一平滑。这有点不自然。而贝叶斯方法，通过利用参数先验服从均匀分布达到相同的效果

边缘似然

- 称$p(\mathcal{D})$ 为边缘似然 (Marginal likelihood)，或证据 (evidence)
  - 在某些时候需要计算出 $p(\mathcal{D})$ 。一般而言，边缘似然非常难计算。但在特殊情况下，还是可以直接计算的，比如在Beta-Bernoulli模型下

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})}$$

$$= \frac{1}{p(\mathcal{D})}\theta^{N_1}(1-\theta)^{N_0}\frac{1}{\mathrm{B}(\alpha,\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}$$

$$= \frac{1}{p(\mathcal{D})}\frac{1}{\mathrm{B}(\alpha,\beta)}\theta^{N_1+\alpha-1}(1-\theta)^{N_0+\beta-1}$$

$$\int p(\theta|\mathcal{D})\, d\theta = 1 \Rightarrow p(\mathcal{D}) = \frac{\mathrm{B}(N_1+\alpha, N_0+\beta)}{\mathrm{B}(\alpha,\beta)}$$

Dirichlet-categorical模型

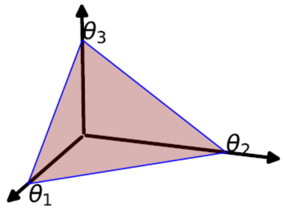
- 由于类型分布是简单的多项分布，故Dirichlet-categorical模型结果很容易推广到Dirichlet-multinomial模型
- 把二值变量结果推广到K值变量。令$x \sim \mathrm{Cat}(\theta)$，即 $p(x) = \mathrm{Cat}(x;\boldsymbol{\theta}) = \prod_{k=1}^{K} \theta_k^{\mathbb{I}(x=k)}$
- 数据集$\mathcal{D} = \{x_n\}_{n=1}^{N}(x_n \in \{1,\dots,K\})$，其似然函数

$$p(\mathcal{D}|\boldsymbol{\theta}) = \prod_{n=1}^{N}\prod_{k=1}^{K}\theta_k^{\mathbb{I}(x=k)} = \prod_{k=1}^{K}\theta_k^{N_k} \qquad N_k \triangleq |\{x_n|x_n = k\}|$$

- 先验分布
  - 为了保证与上述似然函数共轭，θ的先验分布应该具有形式： $p(\theta) \propto \prod_{k=1}^{K}\theta_k^{N_k}$
  - 狄利克雷 (Dirichlet)分布具备上述形式，记$\boldsymbol{\theta} \sim \mathrm{Dir}(\boldsymbol{\alpha})$

$$p(\boldsymbol{\theta}) = \mathrm{Dir}(\boldsymbol{\theta};\boldsymbol{\alpha}) = \frac{1}{\mathrm{B}(\boldsymbol{\alpha})}\prod_{k=1}^{K}\theta_k^{\alpha_k-1} \qquad \boldsymbol{\alpha} \triangleq \{\alpha_1,\dots,\alpha_K\}, \alpha_k > 0 \quad \mathrm{B}(\boldsymbol{\alpha}) = \frac{\prod_{k=1}^{K}\Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^{K}\alpha_k)}$$

Dirichlet-categorical模型



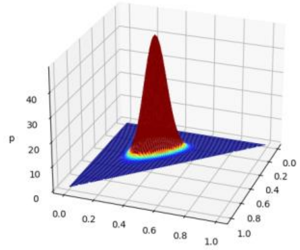
- 狄利克雷 (Dirichlet)分布
  - 是Beta分布的多元变量扩展，也称为关于概率分布的分布
  - 定义域为$\{\boldsymbol{\theta} | 0 \leq \theta_k \leq 1, \sum_{k=1}^{K} \theta_k = 1\}$，即其输入变量是概率分布

$$p(\boldsymbol{\theta}) = \mathrm{Dir}(\boldsymbol{\theta}; \boldsymbol{\alpha}) = \frac{1}{\mathrm{B}(\boldsymbol{\alpha})} \prod_{k=1}^{K} \theta_k^{\alpha_k - 1}$$

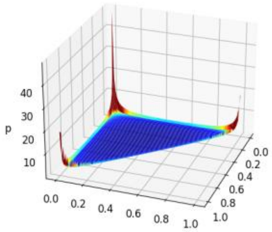
  - 基本性质

均值：$\mathbb{E}[\boldsymbol{\theta}] \triangleq \left[\mathbb{E}[\theta_1], \dots, \mathbb{E}[\theta_K]\right]^{\mathrm{T}}$，$\mathbb{E}[\theta_k] = \frac{\alpha_k}{\sum_{i=1}^{K} \alpha_i}$

众数：$\mathrm{M}[\boldsymbol{\theta}] \triangleq \left[\mathrm{M}[\theta_1], \dots, \mathrm{M}[\theta_K]\right]^{\mathrm{T}}, \mathrm{M}[\theta_k] = \frac{\alpha_k - 1}{\sum_{i=1}^{K} \alpha_i - K}$



$\boldsymbol{\alpha} = \{20, 20, 20\}$



$\boldsymbol{\alpha} = \{0.1, 0.1, 0.1\}$

- 后验分布

$$p(\boldsymbol{\theta}) = \mathrm{Dir}(\boldsymbol{\theta};\boldsymbol{\alpha}) = \frac{1}{\mathrm{B}(\boldsymbol{\alpha})}\prod_{k=1}^{K}\theta_k^{\alpha_k-1}$$

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})} \propto p(\mathcal{D}|\theta)p(\theta)$$

$$\propto \prod_{k=1}^{K}\theta_k^{N_k}\prod_{k=1}^{K}\theta_k^{\alpha_k-1} = \prod_{k=1}^{K}\theta_k^{N_k+\alpha_k-1}$$

$$\propto \mathrm{Dir}(\boldsymbol{\theta};\hat{\boldsymbol{\alpha}}) \quad \hat{\boldsymbol{\alpha}} = \{N_1+\alpha_1,\dots,N_K+\alpha_K\}$$

$$\left.\begin{aligned} p(\theta|\mathcal{D}) &= C\times\mathrm{Dir}(\boldsymbol{\theta};\hat{\boldsymbol{\alpha}}) \\ \int p(\theta|\mathcal{D})d\theta &= 1 \\ \int \mathrm{Dir}(\boldsymbol{\theta};\hat{\boldsymbol{\alpha}})d\theta &= 1 \end{aligned}\right\} \Rightarrow p(\theta|\mathcal{D}) = \mathrm{Dir}(\boldsymbol{\theta};\hat{\boldsymbol{\alpha}}) = \frac{1}{\mathrm{B}(\hat{\boldsymbol{\alpha}})}\prod_{k=1}^{K}\theta_k^{N_k+\alpha_k-1}$$

后验分布统计量

$$p(\theta|\mathcal{D}) = \mathrm{Dir}(\boldsymbol{\theta}; \hat{\boldsymbol{\alpha}}) = \frac{1}{\mathrm{B}(\hat{\boldsymbol{\alpha}})} \prod_{k=1}^{K} \theta_k^{N_k+\alpha_k-1} \qquad \hat{\boldsymbol{\alpha}} = \{N_1+\alpha_1, \ldots, N_K+\alpha_K\}$$

- 均值

$$\mathbb{E}[\boldsymbol{\theta}|\mathcal{D}] \triangleq \left[\mathbb{E}[\theta_1|\mathcal{D}], \ldots, \mathbb{E}[\theta_K|\mathcal{D}]\right]^{\mathrm{T}}, \mathbb{E}[\theta_k|\mathcal{D}] = \frac{N_k+\alpha_k}{\sum_{i=1}^{K}(N_i+\alpha_i)} = \frac{N_k+\alpha_k}{\sum_{i=1}^{K} N_i + \sum_{i=1}^{K} \alpha_i}$$

- 众数

最大后验估计

$$\mathrm{M}[\boldsymbol{\theta}|\mathcal{D}] \triangleq \left[\mathrm{M}[\theta_1|\mathcal{D}], \ldots, \mathrm{M}[\theta_K|\mathcal{D}]\right]^{\mathrm{T}}$$

$$\mathrm{M}[\theta_k|\mathcal{D}] = \frac{N_k+\alpha_k-1}{\sum_{i=1}^{K}(N_i+\alpha_i)-K} = \frac{N_k+\alpha_k-1}{\sum_{i=1}^{K} N_i + \sum_{i=1}^{K} \alpha_i - K}$$

当$\alpha_1 = \alpha_2 = \cdots = \alpha_K = 1$， $\mathrm{M}[\theta_k|\mathcal{D}] = \frac{N_k}{\sum_{i=1}^{K} N_i}$ 最大似然估计

后验预测

$$p(\theta|\mathcal{D}) = \mathrm{Dir}(\boldsymbol{\theta};\hat{\boldsymbol{\alpha}}) = \frac{1}{\mathrm{B}(\hat{\boldsymbol{\alpha}})}\prod_{k=1}^{K}\theta_k^{N_k+\alpha_k-1}$$

$$\begin{aligned} p(x=k|\mathcal{D}) &= \int p(x=k,\theta|\mathcal{D})\,d\theta \\ &= \int p(x=k|\theta,\mathcal{D})p(\theta|\mathcal{D})\,d\theta \\ &= \int p(x=k|\theta)p(\theta|\mathcal{D})\,d\theta \\ &= \int \theta_k \mathrm{Dir}(\boldsymbol{\theta};\hat{\boldsymbol{\alpha}})\,d\theta \\ &= \mathbb{E}[\theta_k|\mathcal{D}] = \frac{N_k+\alpha_k}{\sum_{i=1}^K N_i + \sum_{i=1}^K \alpha_i} \end{aligned}$$

$$\Longrightarrow \quad p(x|\mathcal{D}) = \mathrm{Cat}(x;\hat{\boldsymbol{\alpha}})$$

边缘似然

$$p(\boldsymbol{\theta}) = \frac{1}{\mathrm{B}(\boldsymbol{\alpha})} \prod_{k=1}^{K} \theta_k^{\alpha_k - 1}$$

$$p(\mathcal{D}|\boldsymbol{\theta}) = \prod_{k=1}^{K} \theta_k^{N_k}$$

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})}$$

$$= \frac{1}{p(\mathcal{D})} \prod_{k=1}^{K} \theta_k^{N_k} \frac{1}{\mathrm{B}(\boldsymbol{\alpha})} \prod_{k=1}^{K} \theta_k^{\alpha_k - 1}$$

$$= \frac{1}{p(\mathcal{D})} \frac{1}{\mathrm{B}(\boldsymbol{\alpha})} \prod_{k=1}^{K} \theta_k^{N_k + \alpha_k - 1}$$

$$\int p(\theta|\mathcal{D})\, d\theta = 1 \Rightarrow p(\mathcal{D}) = \frac{\mathrm{B}(\hat{\boldsymbol{\alpha}})}{\mathrm{B}(\boldsymbol{\alpha})} = \frac{\Gamma\left(\sum_{k=1}^{K} \alpha_k\right)}{\prod_{k=1}^{K} \Gamma(\alpha_k)} \times \frac{\prod_{k=1}^{K} \Gamma(N_k + \alpha_k)}{\Gamma\left(\sum_{k=1}^{K} N_k + \sum_{k=1}^{K} \alpha_k\right)}$$

Gaussian-Gaussian模型

- 为论述方便，我们以方差已知情况下的一元高斯分布为例，阐述Gaussian-Gaussian模型基本思想和方法
  - 假设$x \sim \mathcal{N}(\mu, \sigma^2)$，$\mu$未知，$\sigma^2$是一已知常量，$\boldsymbol{\theta} = \mu$
- 数据集$\mathcal{D} = \{x_n\}_{n=1}^N (x_n \in \mathbb{R})$，其似然函数

$$p(\mathcal{D}|\mu, \sigma^2) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x_n - \mu)^2\right) \propto \exp\left(-\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n - \mu)^2\right)$$

- 先验分布
  - 为了保证与上述似然函数共轭，μ的先验分布需具有以上形式。高斯分布满足要求，记

$$\mu \sim \mathcal{N}(\mu_1, \sigma_1^2) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{1}{2\sigma_1^2}(\mu - \mu_1)^2\right)$$

后验分布

$$p(\mu|\mathcal{D}) = \frac{p(\mathcal{D}|\mu)p(\mu)}{p(\mathcal{D})} \propto p(\mathcal{D}|\mu)p(\mu)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2\right)\exp\left(-\frac{1}{2\sigma_1^2}(\mu-\mu_1)^2\right)$$

$$\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2 + \frac{1}{2\sigma_1^2}(\mu-\mu_1)^2 = \frac{1}{2\sigma^2}\left(\boxed{\sum_{n=1}^{N}{x_n}^2} - 2\mu\sum_{n=1}^{N}x_n + \sum_{n=1}^{N}\mu^2\right) + \frac{1}{2\sigma_1^2}(\mu^2 - 2\mu\mu_1 + {\mu_1}^2)$$

$$\bar{x} \triangleq \frac{1}{N}\sum_{n=1}^{N}x_n$$

$$= \frac{1}{2\sigma^2}\left(\boxed{N^2\bar{x}^2 - \sum_{i\neq j}x_i x_j} - 2N\mu\bar{x} + N\mu^2\right) + \frac{1}{2\sigma_1^2}(\mu^2 - 2\mu\mu_1 + {\mu_1}^2)$$

后验分布

$$\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2+\frac{1}{2\sigma_1^2}(\mu-\mu_1)^2=\frac{1}{2\sigma^2}\left(N^2\bar{x}^2-\sum_{i\neq j}x_ix_j-2N\mu\bar{x}+N\mu^2\right)+\frac{1}{2\sigma_1^2}(\mu^2-2\mu\mu_1+{\mu_1}^2)$$

$$=\left(\frac{N}{2\sigma^2}+\frac{1}{2\sigma_1^2}\right)\mu^2-\left(\frac{N\bar{x}}{\sigma^2}+\frac{\mu_1}{\sigma_1^2}\right)\mu+\frac{1}{2\sigma^2}\left(N^2\bar{x}^2-\sum_{i\neq j}x_ix_j\right)+\frac{{\mu_1}^2}{2\sigma_1^2}$$

$$\propto\left(\frac{N}{2\sigma^2}+\frac{1}{2\sigma_1^2}\right)\mu^2-\left(\frac{N\bar{x}}{\sigma^2}+\frac{\mu_1}{\sigma_1^2}\right)\mu$$

$$\Downarrow$$

$$p(\mu|\mathcal{D})\propto\exp\left(-\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2\right)\exp\left(-\frac{1}{2\sigma_1^2}(\mu-\mu_1)^2\right)\propto\exp\left(-\left(\frac{N}{2\sigma^2}+\frac{1}{2\sigma_1^2}\right)\mu^2+\left(\frac{N\bar{x}}{\sigma^2}+\frac{\mu_1}{\sigma_1^2}\right)\mu\right)$$

后验分布

$$p(\mu|\mathcal{D}) \propto \exp\left(-\left(\frac{N}{2\sigma^2}+\frac{1}{2\sigma_1^2}\right)\mu^2+\left(\frac{N\bar{x}}{\sigma^2}+\frac{\mu_1}{\sigma_1^2}\right)\mu\right)$$

$$\alpha \triangleq \frac{N\bar{x}}{\sigma^2}+\frac{\mu_1}{\sigma_1^2} \qquad \beta^2 \triangleq \left(\frac{N}{\sigma^2}+\frac{1}{\sigma_1^2}\right)^{-1}$$

$$p(\mu|\mathcal{D}) \propto \exp\left(-\frac{1}{2\beta^2}(\mu^2-2\beta^2\alpha\mu)\right)$$

$$\propto \exp\left(-\frac{1}{2\beta^2}(\mu^2-2\beta^2\alpha\mu+\beta^4\alpha^2)\right)=\exp\left(-\frac{1}{2\beta^2}(\mu-\beta^2\alpha)^2\right)$$

$$\left.\begin{array}{r} p(\mu|\mathcal{D}) = C\times\mathcal{N}(\mu;\beta^2\alpha,\beta^2) \\ \int p(\mu|\mathcal{D})d\mu = 1 \\ \int \mathcal{N}(\mu;\beta^2\alpha,\beta^2)d\mu = 1 \end{array}\right\} \Rightarrow p(\mu|\mathcal{D}) = \mathcal{N}(\mu;\beta^2\alpha,\beta^2)$$

ITML 后验分布统计量

$$p(\mu|\mathcal{D}) = \mathcal{N}(\mu;\beta^2\alpha,\beta^2) \qquad \alpha \triangleq \frac{N\bar{x}}{\sigma^2} + \frac{\mu_1}{\sigma_1^2} \qquad \beta^2 \triangleq \left(\frac{N}{\sigma^2} + \frac{1}{\sigma_1^2}\right)^{-1}$$

$$\mathbb{E}[\mu|\mathcal{D}] \triangleq \beta^2\alpha = \left(\frac{N}{\sigma^2} + \frac{1}{\sigma_1^2}\right)^{-1}\left(\frac{N\bar{x}}{\sigma^2} + \frac{\mu_1}{\sigma_1^2}\right) = \frac{N\sigma_1^2\bar{x} + \sigma^2\mu_1}{N\sigma_1^2 + \sigma^2} = \frac{N\sigma_1^2}{N\sigma_1^2 + \sigma^2}\bar{x} + \frac{\sigma^2}{N\sigma_1^2 + \sigma^2}\mu_1$$

当仅有一个观测数据，即$\mathcal{D} = \{x\}$时，

$$\mathbb{E}[\mu|\mathcal{D}] = \frac{\sigma_1^2}{\sigma_1^2 + \sigma^2}x + \frac{\sigma^2}{\sigma_1^2 + \sigma^2}\mu_1 = \left(1 - \frac{\sigma^2}{\sigma_1^2 + \sigma^2}\right)x + \frac{\sigma^2}{\sigma_1^2 + \sigma^2}\mu_1 = x - \frac{\sigma^2}{\sigma_1^2 + \sigma^2}(x - \mu_1)$$

由于后验分布$p(\mu|\mathcal{D})$是高斯分布，所以其众数就是均值$\mathbb{E}[\mu|\mathcal{D}]$

方差 $\mathbb{V}[\mu|\mathcal{D}] = \beta^2 = \left(\frac{N}{\sigma^2} + \frac{1}{\sigma_1^2}\right)^{-1} = \frac{\sigma_1^2\sigma^2}{N\sigma_1^2 + \sigma^2}$

后验预测

$$p(\mu|\mathcal{D}) = \mathcal{N}(\mu;\beta^2\alpha,\beta^2)$$

$$p(x|\mu,\mathcal{D}) = p(x|\mu) = \mathcal{N}(x;\mu,\sigma^2)$$

$$\begin{aligned}
p(x|\mathcal{D}) &= \int p(x,\mu|\mathcal{D})\,d\mu \\
&= \int p(x|\mu,\mathcal{D})p(\mu|\mathcal{D})\,d\mu \\
&= \int \mathcal{N}(x;\mu,\sigma^2)\mathcal{N}(\mu;\beta^2\alpha,\beta^2)\,d\mu \\
&= \frac{1}{\sqrt{2\pi\sigma^2}}\frac{1}{\sqrt{2\pi\beta^2}}\int \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)\exp\left(-\frac{1}{2\beta^2}(\mu-\beta^2\alpha)^2\right)d\mu \\
&= \frac{\exp\left(\frac{\hat{\sigma}^2}{2}\left(\frac{x}{\sigma^2}+\alpha\right)^2-\frac{x^2}{2\sigma^2}-\frac{1}{2}\beta^2\alpha^2\right)}{\sqrt{2\pi\sigma^2}\sqrt{2\pi\beta^2}}\frac{\sqrt{2\pi\hat{\sigma}^2}}{\sqrt{2\pi\hat{\sigma}^2}}\int \mathcal{N}\left(\mu;\hat{\sigma}^2\left(\frac{x}{\sigma^2}+\alpha\right),\hat{\sigma}^2\right)d\mu \\
&= \frac{\exp\left(\frac{\hat{\sigma}^2}{2}\left(\frac{x}{\sigma^2}+\alpha\right)^2-\frac{x^2}{2\sigma^2}-\frac{1}{2}\beta^2\alpha^2\right)}{\sqrt{2\pi\sigma^2}\sqrt{2\pi\beta^2}}\sqrt{2\pi\hat{\sigma}^2}
\end{aligned}$$

$p(x|\mathcal{D})$仍然服从正态分布

$$\hat{\sigma}^2 \triangleq \left(\frac{1}{\sigma^2}+\frac{1}{\beta^2}\right)^{-1}$$

边缘似然

$$p(\mu|\mathcal{D}) = \frac{p(\mathcal{D}|\mu)p(\mu)}{p(\mathcal{D})} = \frac{1}{p(\mathcal{D})}\prod_{n=1}^{N}\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{1}{2\sigma^2}(x_n-\mu)^2\right)\frac{1}{\sqrt{2\pi\sigma_1^2}}\exp\left(-\frac{1}{2\sigma_1^2}(\mu-\mu_1)^2\right)$$

$$= \frac{1}{p(\mathcal{D})}\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \frac{1}{\sqrt{2\pi\sigma_1^2}}\exp\left(-\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2-\frac{1}{2\sigma_1^2}(\mu-\mu_1)^2\right)$$

$$\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2+\frac{1}{2\sigma_1^2}(\mu-\mu_1)^2 = \left(\frac{N}{2\sigma^2}+\frac{1}{2\sigma_1^2}\right)\mu^2-\left(\frac{N\bar{x}}{\sigma^2}+\frac{\mu_1}{\sigma_1^2}\right)\mu+\frac{1}{2\sigma^2}\left(N^2\bar{x}^2-\sum_{i\neq j}x_ix_j\right)+\frac{{\mu_1}^2}{2\sigma_1^2}$$

$$\alpha \triangleq \frac{N\bar{x}}{\sigma^2}+\frac{\mu_1}{\sigma_1^2} \qquad \beta^2 \triangleq \left(\frac{N}{\sigma^2}+\frac{1}{\sigma_1^2}\right)^{-1}$$

$$= \frac{1}{2\beta^2}\mu^2-\alpha\mu+\frac{1}{2\sigma^2}\left(N^2\bar{x}^2-\sum_{i\neq j}x_ix_j\right)+\frac{{\mu_1}^2}{2\sigma_1^2}$$

$$\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2+\frac{1}{2\sigma_1^2}(\mu-\mu_1)^2=\frac{1}{2\beta^2}\mu^2-\alpha\mu+\frac{1}{2\sigma^2}\left(N^2\bar{x}^2-\sum_{i\neq j}x_ix_j\right)+\frac{{\mu_1}^2}{2\sigma_1^2}$$

$$=\frac{1}{2\beta^2}(\mu^2-2\beta^2\alpha\mu+\beta^4\alpha^2)-\frac{\beta^2\alpha^2}{2}+\frac{1}{2\sigma^2}\left(N^2\bar{x}^2-\sum_{i\neq j}x_ix_j\right)+\frac{{\mu_1}^2}{2\sigma_1^2}$$

$$p(\mu|\mathcal{D})=\frac{1}{p(\mathcal{D})}\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N\frac{1}{\sqrt{2\pi\sigma_1^2}}\exp\left(-\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2-\frac{1}{2\sigma_1^2}(\mu-\mu_1)^2\right)$$

$$=\frac{1}{p(\mathcal{D})}\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N\frac{1}{\sqrt{2\pi\sigma_1^2}}\exp\left(\frac{\beta^2\alpha^2}{2}-\frac{1}{2\sigma^2}\left(N^2\bar{x}^2-\sum_{i\neq j}x_ix_j\right)-\frac{{\mu_1}^2}{2\sigma_1^2}\right)\exp\left(-\frac{1}{2\beta^2}(\mu-\beta^2\alpha)^2\right)$$

$$=\frac{1}{p(\mathcal{D})}\times Const\times\exp\left(-\frac{1}{2\beta^2}(\mu-\beta^2\alpha)^2\right)\quad Const\triangleq\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N\frac{1}{\sqrt{2\pi\sigma_1^2}}\exp\left(\frac{\beta^2\alpha^2}{2}-\frac{1}{2\sigma^2}\left(N^2\bar{x}^2-\sum_{i\neq j}x_ix_j\right)-\frac{{\mu_1}^2}{2\sigma_1^2}\right)$$

边缘似然

$$p(\mu|\mathcal{D}) = \frac{1}{p(\mathcal{D})} \times Const \times \exp\left(-\frac{1}{2\beta^2}(\mu - \beta^2\alpha)^2\right)$$

$$= \frac{\sqrt{2\pi\beta^2}}{p(\mathcal{D})} \times Const \times \frac{1}{\sqrt{2\pi\beta^2}} \exp\left(-\frac{1}{2\beta^2}(\mu - \beta^2\alpha)^2\right)$$

$$1 = \int p(\mu|\mathcal{D})\, d\mu = \frac{\sqrt{2\pi\beta^2}}{p(\mathcal{D})} \times Const \times \int \frac{1}{\sqrt{2\pi\beta^2}} \exp\left(-\frac{1}{2\beta^2}(\mu - \beta^2\alpha)^2\right) d\mu$$

$$\Downarrow$$

$$1 = \frac{\sqrt{2\pi\beta^2}}{p(\mathcal{D})} \times Const$$

$$\Downarrow$$

$$p(\mathcal{D}) = \sqrt{2\pi\beta^2} \times Const = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \frac{\sqrt{2\pi\beta^2}}{\sqrt{2\pi\sigma_1^2}} \exp\left(\frac{\beta^2\alpha^2}{2} - \frac{1}{2\sigma^2}\left(N^2\bar{x}^2 - \sum_{i\neq j} x_i x_j\right) - \frac{{\mu_1}^2}{2\sigma_1^2}\right)$$

下一讲

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